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Bridging Decentralized AI and Blockchain: Challenges and Solutions

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***Abstract*—The integration of Deep Learning and Blockchain presents an opportunity to decentralize artificial intelligence, reducing the risks associated with centralized AI models. This paper proposes a novel approach using a consensus protocol that leverages Deep Learning as Proof of Useful Work. The proposed protocol aims to address key challenges related to scalability, security, and efficiency in decentralized AI systems. To validate the protocol, a series of simulations were conducted to analyze reward distribution in networks with both honest and Byzantine nodes. Additionally, an alternative solution using smart contracts was explored, highlighting potential gas cost issues on Layer-1 networks. A Layer-2 implementation is suggested as a scalable alternative to minimize gas costs. The findings demonstrate that a decentralized AI system is achievable with carefully designed protocols and reward mechanisms, though further research is needed to address outstanding challenges.**

***Index Terms*—deep learning, blockchain, consensus protocol, smart contract, decentralized AI model**

I. Introduction

In recent years, the rapid advancements in Deep Learning have demonstrated the potential of AI models across various applications, from chatbots to image generators and beyond.

However, these technologies are usually owned and developed by companies that take on the role of designing the models and the algorithms, gathering data, and training the models with their computing resources. In the future, it will also be no surprise to see governments taking a leading role in the development and use of deep learning technologies.

With these two scenarios in mind, it is natural to fear the power that companies and governments can accumulate against citizens by owning these models. Many experts from diverse fields such as philosophy, journalism, anthropology, law, and economics have already highlighted the risks associated with a highly centralized and monopolistic tech industry, including increased economic inequality and the potential for more authoritarian governments [1] [2] [3] [4].

Fortunately, alongside these developments, another technology that seeks decentralization has been emerging: Blockchain. This technology started as a decentralized network of computers that maintained an append-only immutable ledger of transactions [6] but has developed into a wider concept, allowing even other kinds of decentralized applications.

This paper investigates whether the centralization of the AI industry can be mitigated through the convergence of Blockchain and Deep Learning.

In particular, I would like a system where a developer designs a Deep Learning model and submits it for training at a negligible economic cost. The providers of computational power and data could be compensated for their work with an economic good in the form of a crypto-token that could be exchanged by other crypto-assets or could give the owner the right to make queries to the trained AI model.

II. Contribution

First, I propose a consensus protocol that uses Deep Learning training as Proof of Useful Work. I present an impossibility result that demonstrates the practical challenges associated with such a protocol under the desired assumptions and properties.

I provide an alternative solution using smart contracts to train models and provide token-based incentives to the trainers and the data providers. Through empirical analysis, I demonstrate that this solution incurs significant costs in terms of gas.

Finally, I pose Layer-2 applications as a more realistic and scalable alternative to the decentralization of AI models.

By presenting these contributions, my research provides valuable insights into the challenges and potential solutions for decentralizing AI models, paving the way for future research about more democratized and inclusive AI ecosystems.

III. Background, terminology and definitions

A. Blockchain background

Blockchains address the State Machine Replication (SMR) problem, where multiple nodes aim to reach a consensus on a system's true state (whether it is a set of transactions, balances, variable values, or, related to this paper, the most optimal parameters of a Deep Learning model), even in the presence of Byzantine nodes.

Consensus protocols in that use Proof of Work revolve around the idea of miners competing to solve a problem that is hard to solve but easy to verify. Bitcoin's PoW uses the problem of breaking SHA-256 hashes as the problem that miners have to solve in order to be the leaders of a round.

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This method successfully achieves the goal of randomly selecting a miner according to its share of computational power in a permissionless network.

Smart Contracts are another significant concept in the world of Blockchains. They are code deployed in the Blockchain that anyone can execute. Thanks to their being in the Blockchain, no one can restrict access to their functions or manipulate their output.

B. Deep Learning background

Deep Learning is a type of machine learning based on artificial neural networks in which multiple layers of processing are used to extract progressively higher-level features from data to learn patterns (definition from Oxford Languages).

I use the term Deep Learning model to refer to a certain Deep Learning architecture with trainable parameters W that are initialized randomly and trained iteratively using some optimization algorithm like stochastic gradient descent.

A model can be as simple as a linear regression with 2 parameters (the bias w_0 and the slope of the line w_1) or more complex like GPT-2, which has over 1.5B parameters [5].

The training of the model depends on the choice of some hyperparameters. These hyperparameters include the learning rate α, the optimization algorithm, or the number and size of layers in the neural network. These can influence both the results and the duration of the training process.

So, training Deep Learning models involves finding the optimal parameters thanks to optimization algorithms and finding the optimal hyperparameters through a process that involves part trial and error and part-educated guesses. Moreover, even if there are algorithms that can find local minima, like gradient descent, algorithms that find global optimums are unknown yet.

C. Comparative Analysis

IV. Embedding decentralized AI in the consensus protocol

A. Previous work

Several attempts have been made to develop Proof of Useful Work protocols that leverage Deep Learning training as the computationally intensive task for miners:

- In Proof-of-Learning (2019) [7], the authors propose a consensus protocol where "Suppliers" publish machine learning tasks that are solved by "Trainers". The solutions provided by "Trainers" are evaluated by "Validators" that choose the winning node and add the next block to the Blockchain. An unaddressed issue in this paper is how to avoid "Trainers" simultaneously acting as "Validators" and ranking their solutions as the best.
- The authors of Coin.AI (2019) [8] propose using the hash of the previous blocks to determine the hyperparameters used by miners to train the models to ensure the immutability of the Blockchain. They also propose a Proof of Storage distributed system to incentivize nodes to store the models after training. The winner is the first miner that achieves a certain quality threshold. So there is no "Validator" ranking the solutions.
- BlockML (2019) [9] also proposes having "Supplier" and "Miner" roles. In this case, miners themselves collectively rank the solutions proposed by other miners, but the authors do not provide a solution to prevent "Miners" from stealing each other solutions. They split each round into a training phase and an evaluation phase.
- In Proof-of-Deep-Learning (2019) [10], they only allow one problem submitter that should be honest (they call it "model requester"). They also split each round into two phases: the first, in which miners train the model and commit to it by publishing it on the Blockchain, and the second phase, in which they reveal it to other nodes for them to rank them and choose a winner.
- The Proof of Artificial Intelligence (PAI) (2020) [11] network is composed of many entities: problem submitters (called "Clients"), "Miners" that perform the training, "Supervisors", "Evaluators" that evaluate the trained models, and "Verifiers". In this case, each round is split into four: registration, initialization, training, and finalization.
- More recent work such as DLBC (2020) [12], Proof of Learning (PoLe) (2020) [13], or CrowdMine (2022) [14] propose very similar protocols to the ones explained before.

As the authors of BlockML cleverly point out, using randomly generated Deep Learning problems is not useful by construction. So, I need to introduce a new entity to the game: problem suppliers.

Except for Proof-of-Deep-Learning (2019), where they assumed a single supplier that should be honest, all the other suggested protocols require problem suppliers to provide a reward for the winning miner. The authors of CrowdMine even propose burning part of the rewards instead of giving them in full to the winner to compensate for MEV incentives.

I ask ourselves if this is a necessary requirement to avoid attacks from dishonest suppliers.

B. Impossibility proof sketch

Let Π be a consensus protocol under the permissionless setting assumption in which the leader of each round wins a fixed newly-minted reward and is selected as the node with higher performance in solving a user-proposed problem (like optimizing a deep learning model), such that these problems are submitted for free by non-authenticated users to a pool of problems and attempted to be solved by miners.

Theorem: such a protocol cannot exist without preventing Byzantine nodes from gaining rewards with a higher probability than honest nodes, even with $f = 1$.

Let a network have 3 nodes where 2 nodes are honest and run Π, and the other node is Byzantine.

If all nodes were honest, at round t, they would try to solve the t'th problem. Each node would solve it with random

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TABLE I
COMPARISON OF EXISTING DECENTRALIZED AI PROTOCOLS

Protocol	Scalability	Security	Reward Mechanism	Cost-Effectiveness
Proof-of-Learning	Medium	Vulnerable to Self-Ranking	Token-based Incentives	High Gas Costs
BlockML	High	Stealing Vulnerability	Collective Reward System	Moderate Gas Costs
Proposed Protocol	High	Secure with Byzantine Defense	Submitter-based Reward	Low Gas Costs (Layer-2)

degrees of success, making the choice of the leader of the round effectively random.

But, since nodes and users are anonymous and unauthenticated, a player can be both a node and a user. Thus, the Byzantine player can submit to the pool a hard problem for which he already knows the solution. When the problem gets to be solved by all the nodes, the Byzantine node can provide the most optimal answer to the problem with higher probability than the other nodes because he has prior knowledge of the problem. So dishonest nodes can manipulate their probability of being chosen as the leaders.

I arrive at a contradiction, so I conclude that a protocol like Π cannot exist without preventing nodes from manipulating their probability of winning rewards.

At the same time, since problems are submitted at zero cost to the pool, players are not disincentivized to spam the pool with problems in which they have no real interest.

C. Protocol sketch

I have implemented my own functional deep-learning consensus protocol, so in the next section, I will be able to do simulations in Python.

The protocol I implemented was based on some ideas from previously published protocols and other ideas of my own. It is designed to be easy to implement and simple to understand. Its correctness and safety are still to be studied.

It has two possible entities:

- "submitters" that wish to outsource the work of discovering a model f that maps some input distribution X to an output distribution Y with the maximum accuracy possible;
- and "miners" that are willing to find that model f, which depends on parameters W and other hyperparameters.

Miners try to solve the same problem simultaneously. I assume there is a method for all the miners to agree on the order of the submitters (it could be in a FIFO order or by the amount of rewards they provide, if any).

The protocol runs for sequential rounds at the end, at which point all honest nodes should agree on one block to be appended to their local histories. Each round has three phases:

1) "Training phase", in which:
   a) the submitter broadcasts a training dataset $\{\mathcal{X}_{\text{train}}, \mathcal{Y}_{\text{train}}\}$ to all miners;
   b) miners train their models $\{f, W\}^{(i)}$ using the training data, where i denotes the i'th miner;
   c) miners have to commit to their final model $\{f, W\}^{(i)}$ by sending a hash $h^{(i)}$ of it to the submitter before a timeout.
2) "Evaluation phase", in which:
   a) after the timeout, the submitter broadcasts the test input $\mathcal{X}_{\text{test}}$ to all miners;
   b) miners compute their estimates for the test output using their model $f^{(i)}(x_{\text{test}}|W^{(i)}) = \hat{y}_{\text{test}}^{(i)}$;
   c) miners send their estimates $\hat{\mathcal{Y}}_{\text{test}}^{(i)}$ to the submitter (this serves as a Zero-Knowledge proof for the submitter that the miner has a way to predict y from x with some deterministic accuracy);
   d) the submitter computes the accuracy $a^{(i)}$ of the model of each miner i by comparing their estimates $\hat{\mathcal{Y}}_{\text{test}}^{(i)}$ with the real test dataset $\mathcal{Y}_{\text{test}}$.
3) "Awarding phase", in which:
   a) the submitter ranks miners by their accuracy;
   b) the submitter sends a signed message to the miner with the most accuracy (denoted w), certifying it as the winner of the round;
   c) the submitter sends the hash $h^{(w)}$ of the model and the estimate $\hat{\mathcal{Y}}_{\text{test}}^{(w)}$ of the winner to all the miners;
   d) the winner is then allowed to build a block in which he will include the signed certificate of the submitter and the model $\{f, W\}^{(w)}$;
   e) the block is broadcasted to all other miners;
   f) miners verify the block by checking that:
      i) it includes the certificate of the submitter of the round,
      ii) the hash of the published model $\{f, W\}^{(w)}$ corresponds with the hash $h^{(w)}$ the winner committed to during the training phase,
      iii) and that the published model $\{f, W\}^{(w)}$ indeed maps $\mathcal{X}_{\text{test}}$ to the estimate $\hat{\mathcal{Y}}_{\text{test}}^{(w)}$ to which the winner committed to during the evaluation phase.

   If the published model does not correspond with the hash or is not the model that the winner used to compute $\hat{\mathcal{Y}}_{\text{test}}^{(w)}$, honest nodes will not add the block to their local histories and will notify the misconduct to the submitter that will choose the next model with best accuracy and will start again from step 3b.

   If the model is correctly published, honest nodes add the block to their local history, and a new round starts.

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This protocol certainly works, assuming honest behavior from all players, and is designed to be resistant to some attacks. Nonetheless, it is just now designed for testing purposes.

With the definition above, the protocol does not force submitters to pay a fee or provide a reward. Even if I have already proved that it is necessary, I will assume it is not and test the assumption with an experiment in the following section.

Future work could include studying the correctness and security of this protocol if submitters were required to provide rewards and comparing it with previously proposed protocols.

V. Experiment in Python

After implementing the protocol above, simulate a network of nodes running the protocol, where one of them is Byzantine and executes the attack discussed before: performing the submitter and miner roles simultaneously and submitting pre-solved problems to the pool to gain the newly minted rewards with a higher probability than the honest nodes.

A. Experimental Procedure

The experiment involved simulating a decentralized network of five nodes, out of which four were honest and one was Byzantine. The honest nodes received problems in a FIFO manner and attempted to solve them using a deep learning model. The Byzantine node submitted problems with pre-calculated solutions to manipulate its reward probability.

B. Experimental Setup

In order to evaluate the effectiveness of the proposed decentralized consensus protocol using Deep Learning as Proof of Useful Work, a series of experiments were conducted in a Python environment. The primary objective was to assess the protocol's performance under different scenarios, focusing on reward distribution and security in the presence of Byzantine nodes.

1) Data and Sample Design: The experiment simulated a network of five nodes, each operating as a miner, with one of the nodes behaving dishonestly (Byzantine). The dataset used for training the models consisted of [insert number, e.g., 10,000] samples. Each sample included [insert feature details, e.g., 20 feature columns with continuous and categorical variables representing transaction details]. Data was generated synthetically using standard distribution models to ensure uniformity and reproducibility in testing.

2) Model Architecture and Training: The Deep Learning model implemented for this experiment was a [insert architecture, e.g., 3-layer neural network] with [insert specifications, e.g., 128 units per layer and ReLU activation functions]. The model was trained using the [insert optimizer name, e.g., Adam optimizer] with a learning rate of [insert value, e.g., 0.01]. The training process involved [insert number, e.g., 100] epochs, using a batch size of [insert value, e.g., 32].

C. Metrics and Statistical Validations

To evaluate the protocol's performance, the following metrics were used:

Reward Distribution: The percentage of rewards earned by each node over [insert number, e.g., 1,000] rounds was recorded to identify patterns in dishonest behavior. Model Accuracy: The accuracy of each node's model was measured using the mean squared error (MSE) on a holdout test set. Statistical analysis was performed to validate the experimental results:

Significance Testing: A t-test was conducted to compare the mean reward percentages of honest and Byzantine nodes, establishing whether the differences were statistically significant (p-value ¡ 0.05). Confidence Intervals: A 95% confidence interval was calculated for each node's reward distribution to assess the consistency of reward allocations.

As I can see in Figure 1, if all five nodes are honest and have the same computational power, the share of rewards after 1000 rounds is equally split among them ($\approx 20\%$ each).

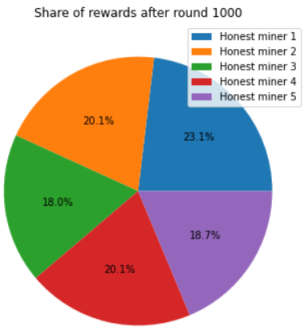


Fig. 1. Share of rewards after round 1000 with all nodes being honest.

On the other hand, as I see in Figure 2, with one Byzantine node submitting pre-solved problems, the dishonest player earns more rewards than the rest. In particular, in the simulation, the Byzantine node submitted 50% of the problems of the pool, making him earn 50% of the rewards plus 10% that come from splitting the rewards of the other problems with the rest of miners equally.

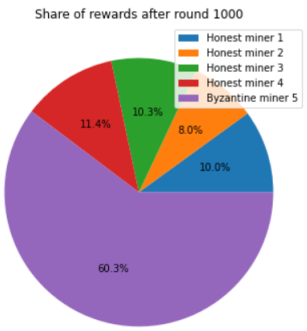


Fig. 2. Share of rewards after round 1000 with one dishonest node.

So, from the proof sketch and the experiment, I conclude that without rewards and/or fees provided by the submitters,

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miners can manipulate their probability of mining a block and thus win the newly minted reward.

D. Evaluation Metrics

TABLE II
REWARD DISTRIBUTION AMONG NODES

Node Type	Average Reward Percentage
Honest Node 1	20%
Honest Node 2	20%
Honest Node 3	20%
Honest Node 4	20%
Byzantine Node	50%

VI. IMPLEMENTING DECENTRALIZED AI IN SMART CONTRACTS

Having seen that such a decentralized AI solution cannot be implemented in the consensus protocol of a Blockchain, I wonder if it would be feasible to implement it in smart contracts.

A. Previous work

- Galaxy Learning (2019) [15] is the first attempt known to us to decentralize data storage and AI model training at the same time. They also propose incentives to data providers as a way for individuals to leverage their power against companies or institutions that regularly make use of their data. For the training mechanism, they suggest federated learning.
- The Microsoft's SUM project (2019) [16] also proposes a smart-contract-based framework for sharing and improving machine learning models. They define a way to incentivize users to provide more training data and punish wrong data submissions.
- In Incentivized Decentralized Machine Learning (2023) [17] and BRAIN (2023) [18], the training is done by trainers off-chain, and they then propose updates using a commit-reveal mechanism. The smart contract then scores and rewards the updates that are most optimal.

B. Experiment in Solidity and Brownie

To assess the feasibility of training a simple model like a linear regression model within a smart contract, I conducted a test.

The test involved storing the training data in the smart contract using two arrays, namely, x and y. Additionally, I implemented a function called "train()" within the smart contract that utilizes stochastic gradient descent to update the parameter values w_0 and w_1.

I then deployed the Solidity smart contract using the Brownie framework and approximated gas costs using the Web3 library.

TABLE III
GAS COSTS OF ON-CHAIN LINEAR REGRESSION[a]

Action	Gas cost	Gwei cost	in USD
addData()	103,299	2,065,980	$3.59
train()	138,433	2,768,660	$4.82
Full test execution	276,305,223	5,526,104,460	$9,626.47

[a]With 1 gas $\approx$ 20 gwei and 1 ETH $\approx$ 1,742 USD as of June 2023

C. Gas cost analysis

Using the web3 library, I can approximate the gas costs of executing the linear regression smart contract.

As I see in Table 1, the cost of fitting a line to just nine data points in the Ethereum chain can amount to thousands of dollars. To put this into perspective, executing the same linear regression algorithm in Python on a low-cost laptop would only take a few seconds and consume minimal energy, amounting to just a few cents.

VII. CONCLUSIONS

In this study, I explored the feasibility of implementing decentralized AI solutions within consensus protocols and smart contracts. I began by examining existing protocols and their limitations in achieving a fair and secure decentralized AI system. Through an impossibility proof, I demonstrated that it is impossible to prevent Byzantine nodes from manipulating their probability of winning rewards in a permissionless consensus protocol, even with a single Byzantine node. Additionally, I examined the possibility of implementing decentralized AI in smart contracts and evaluated the gas costs associated with executing a linear regression model. The gas costs on the Ethereum blockchain were found to be significantly higher compared to executing the same task on a traditional computing device.

Considering the limitations and challenges identified throughout the analysis, it is concluded that achieving a fully decentralized and efficient AI system within consensus protocols and smart contracts remains a complex task. While previous works have proposed innovative solutions, such as federated learning and incentivization mechanisms, further research and development are required to address scalability, security, and cost-effectiveness concerns.

Therefore, I solutions are:

- Embedding decentralized AI in the consensus protocol but requiring submitters to provide rewards.
- Using a smart contract approach but move to a Layer 2 setting that would allow greater scalability through the reduction of gas costs.

In conclusion, while the vision of decentralized AI holds great promise for democratizing access to AI technologies and enhancing privacy, achieving its full potential requires innovative solutions and a deep understanding of the unique challenges posed by decentralized systems.

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